# FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – I

## JEE (Main)-2021

### TEST DATE: 17-02-2021

**Time Allotted: 3 Hours** **Maximum Marks: 300**

**General Instructions:**

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  **Section-A (01 – 20, 31 – 50, 61 – 80)** *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  ***Section-B (21 – 30, 51 – 60, 81 – 90)*** *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

# Physics

## PART – I

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

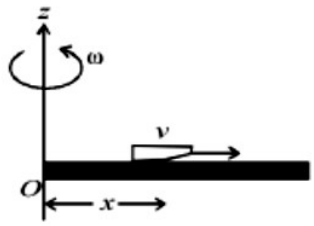
1. The dimensions of $\frac{1}{\epsilon_0}\frac{e^2}{hc}$ are
   (A) $M^{-1}L^{-3}T^4A^2$
   (B) $ML^3T^{-4}A^{-2}$
   (C) $M^0L^0T^0A^0$
   (D) $M^{-1}L^{-3}T^2A$

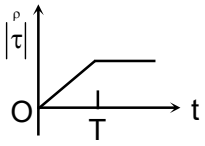
2. A particle moving along x-axis has acceleration f, at time t, given by $f = f_0\left(1-\frac{t}{T}\right)$, where $f_0$ and T are constants. The particle at t = 0 has zero velocity. In the time interval between t = 0 and the instant when f = 0, the particle's velocity ($v_x$) is
   (A) $\frac{1}{2}f_0T^2$
   (B) $f_0T^2$
   (C) $\frac{1}{2}f_0T$
   (D) $f_0T$

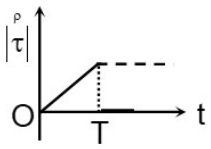
3. A body of mass M is kept on a rough horizontal surface (friction coefficient $\mu$). A person is trying to pull the body by applying a horizontal force but the body is not moving. The force by the surface on the body is F, then
   (A) F = Mg
   (B) F = $\mu$Mg
   (C) $Mg \le F \le Mg\sqrt{1+\mu^2}$
   (D) $Mg \ge F \ge Mg\sqrt{1+\mu^2}$

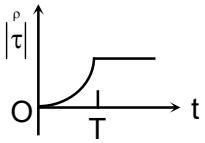
4. A particle falls from a height h on a fixed horizontal plane and rebounds. If e is the coefficient of restitution, the total distance travelled by the particle before it stops rebounding is
   (A) $\frac{h}{2}\frac{[1-e^2]}{[1+e^2]}$
   (B) $\frac{h[1-e^2]}{[1+e^2]}$
   (C) $\frac{h}{2}\frac{[1+e^2]}{[1-e^2]}$
   (D) $\frac{h[1+e^2]}{[1-e^2]}$

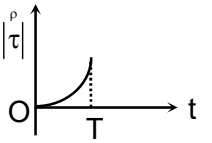
5. A thin uniform rod, pivoted at O, is rotating in the horizontal plane with constant angular speed $\omega$, as shown in the figure. At time t = 0, a small insect starts from O and moves with constant speed v, with respect to the rod towards the other end. It reaches the end of the rod at t = T and stops. The angular speed of the system remains $\omega$ throughout. The magnitude of the torque $(|\vec{\tau}|)$ about O, as a function of time is best represented by plot




(A) 

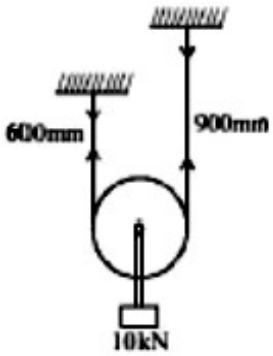

(B) 


(C) 

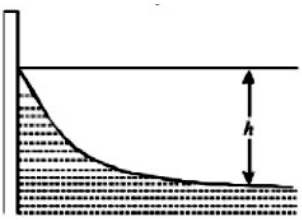

(D) 


6. A load of 10 kN is supported from a pulley which in turn is supported by a rope of sectional area $1 \times 10^3$ mm$^2$ and modulus of elasticity $10^3$ N mm$^{-2}$, as shown in figure. Neglecting the friction at the pulley, determine the deflection of the load.

(A) 2.75 mm
(B) 3.75 mm
(C) 5.25 mm
(D) 6.50 mm




7. Water of density $\rho$ in a clean aquarium forms a meniscus, as illustrated in the figure. Calculate the difference in height h between the centre and the edge of the meniscus. The surface tension of water is $\gamma$.



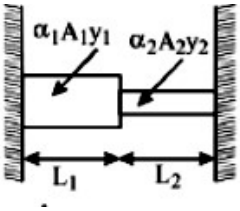

(A) $\sqrt{\frac{2\gamma}{\rho g}}$

(B) $\sqrt{\frac{\gamma}{\rho g}}$

(C) $\frac{1}{2}\sqrt{\frac{\gamma}{\rho g}}$

(D) $2\sqrt{\frac{\gamma}{\rho g}}$

8. Two elastic rods are joined between fixed supports as shown in figure. Condition for no change in the length of individual rods with the increase of temperature. ($\alpha_1$, $\alpha_2$ = linear expansion coefficient, $A_1$, $A_2$ = area of rods, $y_1$, $y_2$ : Youngs modulus)




(A) $\frac{A_1}{A_2} = \frac{\alpha_1}{\alpha_2}\frac{y_1}{y_2}$

(B) $\frac{A_1}{A_2} = \frac{L_1\alpha_1 y_1}{L_2\alpha_2 y_2}$

(C) $\frac{A_1}{A_2} = \frac{L_2\alpha_2 y_2}{L_1\alpha_1 y_1}$

(D) $\frac{A_1}{A_2} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$

9. The heat (Q) supplied to a solid, which is otherwise thermally isolated from its surroundings, is plotted as a function of its absolute temperature, $\theta$. It is found that they are related by the equation.

$Q = a\theta^2 + b\theta^4$ (a, b is constants)

The heat capacity of the solid is given by

(A) $a\frac{\theta^3}{3} + b\frac{\theta^5}{5}$

(B) $a\theta + b\theta^3$

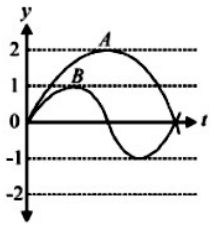
(C) $a\frac{\theta}{3} + b\frac{\theta^3}{5}$

(D) $2a\theta + 4b\theta^3$

10. A particle free to move along the x-axis has potential energy given by $U(x) = k[1 - \exp(-x^2)]$ for $-\infty \le x \le +\infty$, where k is a positive constant of appropriate dimension. Then

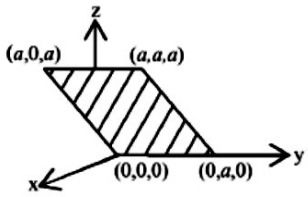
(A) at points away from the origin, the particle is in unstable equilibrium.
(B) for any finite nonzero value of x, there is a force directed away from the origin.
(C) if its total mechanical energy is k/2, it has minimum kinetic energy at the origin.
(D) for small displacements from x = 0, the motion is simple harmonic

11. The displacement-time graphs for two sound waves A and B are shown in the figure. Then the ratio of their intensities $I_A/I_B$ is equal to




(A) 1 : 4
(B) 1 : 16
(C) 1 : 2
(D) 1 : 1

12. Consider an electric field $\vec{E} = E_0\hat{x}$ where $E_0$ is a constant. The flux through the shaded area (as shown in the figure) due to this field is



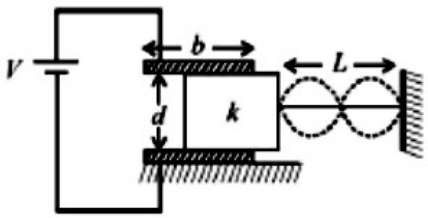

(A) $2E_0a_2$

(B) $\sqrt{2}E_0a^2$

(C) $E_0a^2$

(D) $\frac{E_0a^2}{\sqrt{2}}$

13. A dielectric slab is attached to a string of mass per unit length $\mu$, whose other end is fixed to a wall. Capacitor has square plates of side b and separation between the plates is d. Find the fundamental frequency of vibration of the string. (Dielectric slab remains in equilibrium)




(A) $\frac{1}{2L}\sqrt{\frac{\in_0 bV^2(k-1)\mu}{2d}}$

(B) $\frac{1}{2L}\sqrt{\frac{\in_0 bV^2(k+1)\mu}{2d}}$

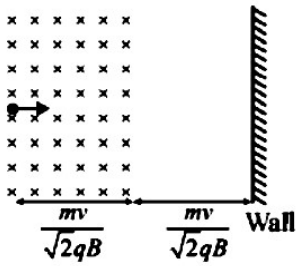
(C) $\frac{1}{2L}\sqrt{\frac{\in_0 bV(k-1)}{2d\mu}}$

(D) $\frac{1}{2L}\sqrt{\frac{\in_0 bV^2(k-1)}{2d\mu}}$

14. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by $\Delta T$ in a time t. A number N of similar cells is now connected in series with a wire of the same material and cross-section but of length 2L. The temperature of the wire is raised by the same amount $\Delta T$ in the same time t. The value of N is

(A) 4
(B) 6
(C) 8
(D) 9

15. A particle of mass m and charge q enters a region of magnetic field (as shown) with speed v. There is a region in which the magnetic field is absent, as shown. The particular after entering the region collides elastically with a rigid wall. Time after which the velocity of particle becomes antiparallel to its initial velocity is



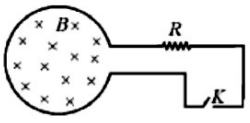

(A) $\frac{m}{2qB}(\pi+4)$

(B) $\frac{m}{qB}(\pi+2)$

(C) $\frac{m}{4qB}(\pi+2)$

(D) $\frac{m}{4qB}(2\pi+3)$

16. Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = B_e e^{-t}$ is established inside the coil. If the key (K) is closed, the electrical power developed right after closing the switch is equal to




(A) $\frac{B_0^2\pi r^2}{R}$

(B) $\frac{B_0 10r^3}{R}$

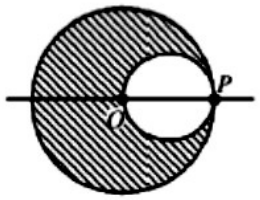
(C) $\frac{B_0^2\pi^2R^4R}{5}$

(D) $\frac{B_0^2\pi^2r^4}{R}$

17. When resonance is produced in a series LCR circuit, then which of the following is not correct?
(A) Current in the circuit is in phase with the applied voltage
(B) Inductive and capacitive reactance's are equal
(C) If R is reduced, the voltage across capacitor will increase
(D) Impedance of the circuit is maximum

18. An electromagnetic wave of frequency $1 \times 10^{14}$ hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\varepsilon_0 = 8.8 \times 10^{-12}$ $C^2/N\text{-}m^2$, then average energy density of electric field will be
(A) $35.2 \times 10^{-10} J/m^3$
(B) $35.2 \times 10^{-11} J/m^3$
(C) $35.2 \times 10^{-12} J/m^3$
(D) $35.2 \times 10^{-13} J/m^3$

19. A transparent sphere of radius R has a cavity of radius R/2 as shown in figure. Find the refractive index of the sphere if a parallel beam of light falling on left surface focuses at point P.




(A) $\mu = \frac{3+\sqrt{5}}{2}$

(B) $\mu = \frac{3-\sqrt{5}}{2}$

(C) $\mu = 3+\sqrt{5}$

(D) $\mu = \frac{1+\sqrt{5}}{2}$

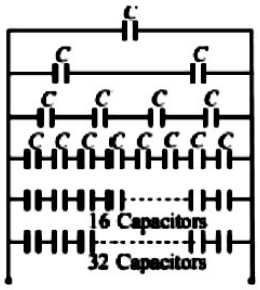
20. The contrast in the fringes in an interference pattern depends on
(A) fringe width
(B) wavelength
(C) intensity ratio of the sources
(D) distance between the slits

## SECTION – B
### (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A metal rod AB of length 10x has its one end A in ice at 0°C, and the other end B in water at 100°C. If a point P on the rod is maintained at 400°C, then it is found that equal amounts of wate and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g. If the point P is at a distance of $\lambda x$ from the ice end A, find the value of $\lambda$.
[Neglect any heat loss to the surrounding].

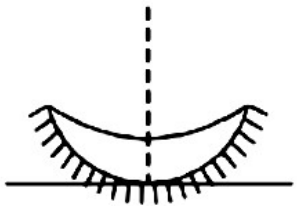
22. Infinite number of identical capacitors (each of capacity 1$\mu$F) are connected as shown in figure. Find the equivalent capacitance (in $\mu$F) of system between the terminals shown in figure.



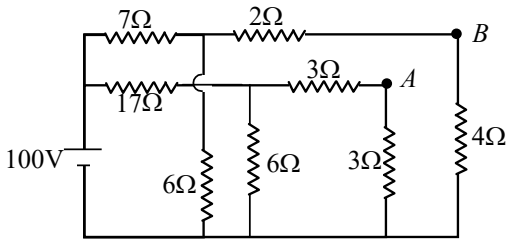

23. In an $\alpha$-decay the kinetic energy of $\alpha$-particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is X. Find the value of X/4. (Assume that daughter nucleus is in ground state)

24. Electrons in hydrogen like atom (Z = 3) make transitions from the fifth to the fourth orbit and from the fourth to the third orbit. The resulting radiations are incident normally on a metal plate and eject photoelectrons. The stopping potential for the photoelectrons ejected by the shorter wavelength is 3.95 volts. Calculate the work function of the metal (in eV).
(Rydberg constant = $1.094 \times 10^7$ $m^{-1}$)

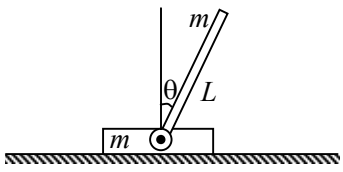
25. The convex surface of a thin concavo-convex lens of glass of refractive index 1.5 has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface. At what distance should a pin be placed on optical axis so that image is formed at same place.

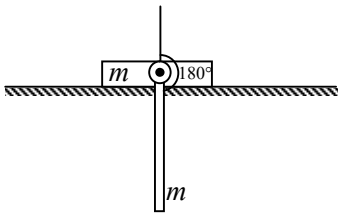


26. Determine the potential difference between points *A* and *B*, $V_{AB}$, in the circuit shown.

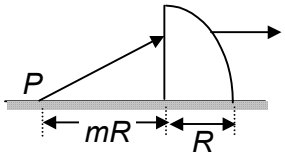



27. A rod of mass *m* and length *L* (= 50/3m) is hinged in plank of same mass *m*. The plank is kept on a smooth horizontal surface and rod makes $\theta$ with the vertical. The system is released from $\theta = 0°$. Find the velocity(in m/s) of plank when the rod makes $\theta = 180°$. ($g$ = 10 m/s$^2$)

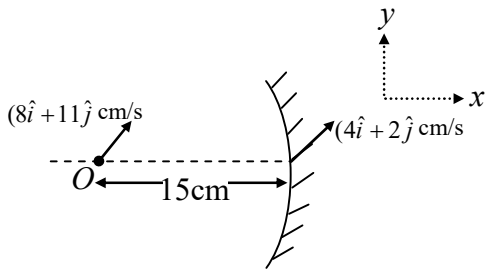



28. A quarter cylinder of glass of radius *R* and refractive index 1.5 is placed on a table. A point object *P* is kept at a distance of *mR* from it. Find the value of *m* for which a ray from *P* will emerge parallel to the table as shown in the figure.



29. A point object located at a distance 15 cm from the pole of a concave mirror of focal length 10 cm on its principal axis is moving with a velocity $(8\hat{i}+11\hat{j})$ cm/s and velocity of mirror is $(4\hat{i}+2\hat{j})$ cm/s as shown. If $\vec{V}$ is the velocity of image. Then find the value of $\left|\vec{V}\right|$ (in cm/s)




30. A drop of water of mass *m* = 0.4 g is placed between two clean glass plates, the distance between which is 0.01 cm. Find the force of attraction between the plates. Surface tension of water = 0.08 N/m.

# Chemistry

## PART – II

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The equilibrium constant expression for the equilibrium
$2NH_3(g) + 2O_2(g) \rightleftharpoons N_2O(g) + 3H_2O(g)$

(A) $K_C = \frac{[N_2O][H_2O]^3}{[NH_3][O_2]}$

(B) $K_C = \frac{[N_2O][H_2O]^3}{[NH_3]^2[O_2]^2}$

(C) $K_C = \frac{[NH_3]^2[O_2]^2}{[N_2O][H_2O]^3}$

(D) $K_C = \frac{[NH_3][O_2]}{[N_2O][H_2O]}$

32. One mole of an ideal monoatomic gas expands isothermally against constant external pressure of 1 atm from initial volume of 1L to a state where its final pressure becomes equal to external pressure. If initial temperature of gas is 300K then total entropy change of system in the above process is:
[R = 0.082L atm $mol^{-1}$ $K^{-1}$ = 8.3 $mol^{-1}$ $K^{-1}$]
(A) 0
(B) Rln(24.6)
(C) Rln(2490)
(D) 3/2Rln(24.6)

33. Arrange the group 15 hydrides in the increasing order of H-M-H bond angle.
(A) $SbH_3 < AsH_3 < PH_3 < NH_3$
(B) $NH_3 < PH_3 < AsH_3 < SbH_3$
(C) $NH_3 < SbH_3 < AsH_3 < PH_3$
(D) $NH_3 < PH_3 < SbH_3 < AsH_3$

34. Identify the correct sequence of increasing number of $\pi$-bonds in structure of following molecules
(I) $H_2S_2O_6$ (II)$H_2SO_3$ (III) $H_2S_2O_5$
(A) I, II, III
(B) II, III, I
(C) II, I, III
(D) I, III, II

35. The inert gas abundantly found in atmosphere is-
(A) Xe
(B) Kr
(C) Ar
(D) He

36. The magnetic moment (B.M.) of $[Fe(CN)_6]^{4-}$ & $[Fe(H_2O)_6]^{2+}$ respectively are-

(A) $\sqrt{24}$.zero

(B) $\sqrt{24}.\sqrt{24}$

(C) zero.$\sqrt{24}$

(D) zero.zero

37. Certain quality of current is passed throught 2 voltameters connected in series and containing $XSO_4(aq)$ and $Y_2SO_4(aq)$ respectively. If the atomic masses of X and Y are in the ratio 2 : 1 the ratio of the masses of Y liberated to that of X is:

(A) 1 : 1

(B) 1 : 2

(C) 2 : 1

(D) 3 : 2

38. A red colouration or precipitate is not obtained when:

(A) $Fe^{3+}$ reacts with potassium thiocyanate

(B) $Fe^{2+}$ reacts with dimethylglyoxime.

(C) $Hg^{2+}$ reacts with potassium iodide.

(D) None of these

39. The shortest wavelength in H spectrum of Lyman series when $R_H = 109678\ cm^{-1}$ is-

(A) $1215.67\,\text{Å}$

(B) $911.7\,\text{Å}$

(C) $1002.7\,\text{Å}$

(D) $1127.30\,\text{Å}$

40. The partial pressure of hydrogen in a flask containing two grams of hydrogen and 32 gm of sulphur dioxide is

(A) 1/16$^{th}$ of the total pressure

(B) 1/9$^{th}$ of the total pressure

(C) 2/3 of the total pressure

(D) 1/8$^{th}$ of the total pressure

41. Arrange the following nucleophiles in the order of their nucleophilic strength:

(A) $OH^- > CH_3COO^- > OCH_3^- > C_6H_5O^-$

(B) $CH_3COO^- < C_6H_5O^- < CH_3O^- < OH^-$

(C) $C_6H_5O^- < CH_3COO^- < CH_3O^- < OH^-$

(D) $CH_3COO^- < C_6H_5O^- < OH^- < CH_3O^-$

42. Hydrogen:

(A) resembles in 1A since it forms mono valent cation H

(B) resembles in VII A (halogen family) since it forms monovalent anion $H^-$

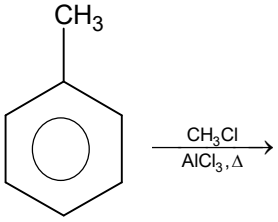
(C) resembles in IV A (carbon family) since both have a half filled shell of electron

(D) All Correct

43. The type (s) of bonds present in diborane is/are
(A) Covalent
(B) One centre bond
(C) Covalent and three centre bond
(D) Covalent and one centre bond

44. In which of the following crossed aldol condensations, only one kind of cross aldol is formed-
(A) $CH_3CHO \& CH_3CH_2CHO$
(B) $CH_3CHO \& (CH_3)_2CO$
(C) $(CH_3)_2CO \& (C_2H_5)_2CO$
(D) $C_6H_5CHO \& CH_3CHO$

45. Major product of this reaction will be-




(A) o-xylene
(B) p-xylene
(C) Both
(D) m-xylene

46. The letter D in D-glucose signifies-
(A) dextrorotatory
(B) mode of synthesis
(C) its configuration
(D) its diamagnetic nature

47. At what $\frac{[Br^-]}{\sqrt{[CO_3^{2-}]}}$ does the following cell have its reaction at equilibrium?

$Ag(s) | Ag_2CO_3(s) | Na_2CO_3(aq) || KBr(aq) | AgBr(s) | Ag(s)$

$K_{sp} = 8 \times 10^{12}$ for $Ag_2CO_3$ and $K_{sp} = 4 \times 10^{12}$ for AgBr

(A) $\sqrt{1} \times 10^{-7}$
(B) $\sqrt{2} \times 10^{-7}$
(C) $\sqrt{3} \times 10^{-7}$
(D) $\sqrt{4} \times 10^{-7}$

48. The shortest wave length in H spectrum of Lyman series when $R_H = 109678\,cm^{-1}$ is –
(A) $1215.67\,\text{Å}$
(B) $911.7\,\text{Å}$
(C) $1002.7\,\text{Å}$
(D) $1127.30\,\text{Å}$

49. What is the pH of 1.0 M $Na_3PO_4$ in aqueous solution?
$PO_4^{3-} + H_2O \rightleftharpoons HPO_4^{2-} + OH^-; K_b = 2.4 \times 10^{-2}$
(A) 13.15
(B) 12.15
(C) 11.15
(D) 10.15

50. Cost of electricity for the production of x L $H_2$ at NTP at cathode is Rs x; then cost of production of x L O, at NTP at anode will be (assume 1 mole of electron is one unit of electricity)
(A) 2x
(B) 4x
(C) 16x
(D) 32x

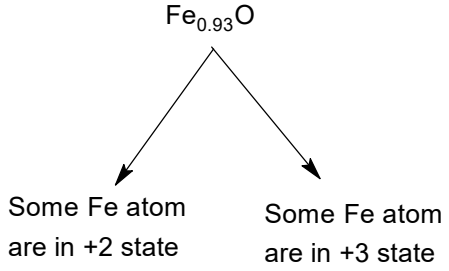
## SECTION – B
### (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. A current of 2.0 A passed for 5 hours through a molten metal salt deposits 22.2g of metal (at wt. = 177). The oxidation state of the metal in the metal salt is:

52. Estimate the difference in energy between 1st and 2nd bohr orbit for a H atom. At what minimum atomic no., a transition from n = 2 to n = 1 energy level would result in the emission of X-ray.with $\lambda = 3.0 \times 10^{-8}$m. Which hydrogen spectrum like species does this atomic no. Corresponds to-

53. Ferrous oxide (FeO) is experimentally found to have the formula $Fe_{0.93}O$. Find the % age of Fe ions in +3 state.




54. Calculate pH of a KOH solution having:
5.6 g of KOH mixed in 50 water

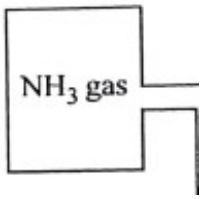
55. Calculate pH of a KOH solution having:
If it is further diluted to make 100 ml.

56. The molality and molarity of a solution of $H_2SO_4$ are 94.13 and 11.12 respectively. Calculate the density of the solution.

57. Assuming that $O_2$ molecule is spherical in shape with radius $2A^o$, the percentage of the volume of $O_2$ molecules to the total volume of gas at S.T.P. is:

58. A manometer attached to a flask contains $NH_3$ gas have no difference in mercury level initially as shown in diagram. After the sparking into the flask, it have difference of 19 cm in mercury level in two columns. Calculate % dissociation of ammonia

have difference of 19 cm in




59. Calculate the amount of ice that will separate out of cooling a solution containing 50g of ethylene glycol in 200 g water to -9.3°C. ($K_f$ for water = 1.86 K $mol^{-1}$ kg)

60. The vapour pressure of an aqueous solution of glucose is 750 mm Hg at 373 K. Calculate molality and mole fraction of solute.

# Mathematics

## PART – III

### SECTION – A
### (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Evaluate $\sum_{n=2}^{\infty} \log_2\left(1-\frac{1}{n^2}\right)$

(A) 0
(B) –1
(C) 1
(D) 2

62. A and B are two non–singular square matrices of each 3 × 3 such that AB = A and BA = B and $|A + B| \neq 0$ then

(A) $|A + B| = 0$
(B) $|A + B| = 8$
(C) $|A – B| = 1$
(D) $|A + B| = 2$

63. For integer n > 1, the digit at units place in the number $\sum_{r=0}^{20} r! + 2^{2^n}$ is

(A) 0
(B) 4
(C) 6
(D) 9

64. If $\sum_{i=1}^{4}\left(x_i^2 + y_i^2\right) \leq 2x_1x_3 + 2x_2x_4 + 2y_2y_3 + 2y_1y_4$ then points $(x_1, y_1)$, $(x_2, y_2)$, $(x_3, y_3)$, $(x_4, y_4)$ are

(A) vertices of a rectangle
(B) collinear
(C) concyclic
(D) none of these

65. If (a cos $\theta_1$, a sin $\theta_1$), (a cos $\theta_2$, a sin $\theta_2$), (a cos $\theta_3$, a sin $\theta_3$) represent the vertices of an equilateral triangle inscribed in a circle, then

(A) $\sum \sec\theta_i = 0$
(B) $\sum \sin\theta_i = 0$
(C) $\sum \tan\theta_i = 0$
(D) $\sum \cot\theta_i = 0$

66. If $y = x\log\left[\frac{x}{a+bx}\right]$ then $x^3\frac{d^2y}{dx^2}$ is equal to

(A) $\left(x\frac{dy}{dx}-y\right)^2$

(B) $\frac{a^2x}{(a+bx)^2}$

(C) $\left(\frac{dy}{dx}-y\right)^2$

(D) $\left(x\frac{dy}{dx}+y\right)^2$

67. If the line x + 2y + 4 = 0, cutting the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ in points whose eccentric angles are 30º and 60º subtends right angle at the origin then its equation is

(A) $\frac{x^2}{8}+\frac{y^2}{4}=1$

(B) $\frac{x^2}{16}+\frac{y^2}{4}=1$

(C) $\frac{x^2}{4}+\frac{y^2}{16}=1$

(D) none of these

68. If the two roots of the equation $(c-1)(x^2+x+1)^2-(c+1)(x^4+x^2+1)=0$ are real and distinct and $f(x)=\frac{1-x}{1+x}$ then $f(f(x))+f\left(f\left(\frac{1}{x}\right)\right)$ is

(A) –c
(B) c
(C) 2c
(D) –2c

69. Let W denotes the set of words in the English dictionary. Define the relation R by R = {(x, y) $\in$ W × W; the words x and y have as least one letter in common}. Then R is

(A) reflexive, symmetric not transitive
(B) reflexive, symmetric and transitive
(C) reflexive, not symmetric, and transitive
(D) Not reflexive, symmetric and transitive

70. If $\alpha$ and $\beta$ are the roots of the equation tan x = 2x; then the value of $\int_{-1}^{1}\sin\alpha x\sin\beta x\,dx(\alpha,\ \beta>0)$ is

(A) 0
(B) 1
(C) 2
(D) 3

71. Let A = {1, 2, 3, 4, 5}. Then the number of one–one onto functions from A to A in which at least three elements have self image, is equal to
(A) 10
(B) 11
(C) 120
(D) $5^5 - 5!$

72. Let f and g be two functions with f″ and g″ existing everywhere. If f(x)g(x) = 2 $\forall$ x and $f'(x), g'(x) \neq 0$. Then $\frac{f''(x)}{f'(x)} = \frac{g''(x)}{g'(x)}$
(A) has at least one real root
(B) may have real roots
(C) has at most one real root
(D) has no real roots

73. If the standard deviation of $x_1, x_2 \ldots\ldots, x_n$ is 3.5, the standard deviation of $-2x_1 - 3, -2x_2 - 3, \ldots\ldots, -2x_n - 3$ is
(A) –7
(B) –4
(C) 7
(D) 1.75

74. Sum of first n terms of an AP (having positive terms) is given by $S_n = (1+2T_n)(1-T_n)$ (where $T_n$ is the nth term of the series). Then the value of $2T_1^2$ is
(A) 2
(B) 4
(C) 1
(D) 6

75. Let f(x) = $\int_0^x (t^2 + 2t + 2)\,dt$, where x is set of real numbers satisfying the inequation $\log_2\left(1+\sqrt{6x - x^2 - 8}\right) \geq 0$. If range of f(x) $\in$ [$\alpha$, $\beta$] then maximum value of $|\alpha - \beta|$ is
(A) $\frac{103}{3}$
(B) $\frac{104}{3}$
(C) $\frac{106}{3}$
(D) none of these

76. An insurance company believes that people can be divided into two classes; those who are accident prone and those who are not. Their statistics show that an accident prone person will not have an accident in a yearly period with probability 0.4 whereas this probability is 0.2 for the other kind. Given that 30% of people are accident prone. The probability that a new policyholder will have an accident within a year of purchasing a policy is
(A) 0.74
(B) 0.28
(C) 0.34
(D) 0.66

77. If $H_1, H_2, H_3, \ldots\ldots, H_{2n+1}$ are in H.P., then $\sum_{i=1}^{2n}(-1)^i\left(\frac{H_i + H_{i+1}}{H_i - H_{i+1}}\right)$ is equal to

(A) 2n – 1
(B) 2n + 1
(C) 2n
(D) 2n + 2

78. The all possible values of a such that
$\sin^2\left(\frac{2\pi}{5}x\right) + a\sin\left(\frac{2\pi}{5}x\right) + \sin^{-1}(x^2 - 10x + 26) + \cos^{-1}(x^2 - 10x + 26) = 0$ has atleast one solution is/are

(A) $\left(-\alpha, -\sqrt{2\pi}\right] \cup \left[\sqrt{2\pi}, \alpha\right)$
(B) $\left(-\alpha, -\sqrt{2\pi}\right] \cup \left(\sqrt{\pi}, \alpha\right)$
(C) $\left(-\sqrt{2\pi}, \sqrt{2\pi}\right)$
(D) none of these

79. Let $z = \begin{vmatrix} \omega^{13} & \omega^5 & \omega^{10} \\ 2i & -3i & 7i \\ \omega^8 & \omega^7 & \omega^{14} \end{vmatrix}$, where $\omega$ is complex cube root of unity then

(A) z is purely real number
(B) z is purely imaginary
(C) z has equal real and imaginary part
(D) none of these

80. The function $f(x) = 2x^3 + \alpha x^2 + \beta x + r$ where $\alpha, \beta, r \in R$ has local minima at $x = 3^t$ and local max at $x = 2\cdot 3^t$. It $x = \frac{5}{2}$ is the point of inflection then the possible value of t is/are

(A) $\log_3 5$
(B) $\log_5 3$
(C) $\log_5 \frac{3}{5}$
(D) $\log_3 \frac{5}{3}$

## SECTION – B
### (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Consider the graph of $y = ax^2$, $a \in R^+$ and $y^2 - x^2 = 4y - 3$ then number of points of intersections is/are

82. The points represented by $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar and
$(\sin A)\vec{a} + (2\sin 2B)\vec{b} + (3\sin 3C)\vec{c} - 4\vec{d} = \vec{0}$ then, the least value of
$\frac{21}{8}(\sin^2 A + \sin^2 2B + \sin^2 3C)$ is

83. In a $\Delta ABC$ angle A = $\frac{2\pi}{3}$ and BC + AC = 20 and AB + BC = 21, then the length of side BC is

84. The number of integral order pair (x, y) that satisfy the system of equation
$|x + y - 4| = 5$ and $|x - 3| + |y - 1| = 5$ is/are

85. The equation of the plane through the intersection of the planes $x + 2y + z - 1 = 0$ and $2x + y + 3z - 2 = 0$ and perpendicular to the plane $x + y + z - 1 = 0$ and $x + ky + 3z - 1 = 0$. Then the value of k is

86. The number of positive integral solutions of the equation $x_1 x_2 x_3 = 60$ is

87. In a triangle ABC if line joining circumcentre P and orthocentre O is parallel to side BC then $\frac{\cos A}{\cos(B - C)}$ is equal to

88. Let $F(x) = f(x).g(x).h(x) \quad \forall \in R$, where f(x), g(x), h(x) are differentiable functions. At a point $x_1$, $F'(x_1) = 21F(x_1)$, $f'(x_1) = 4f(x_1)$, $g'(x_1) = -7g(x_1)$ and $h'(x_1) = kh(x_1)$. Then k is equal to

89. If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide then length of latus rectum of ellipse is

90. The value of $\cot^2 36^\circ \cot^2 72^\circ$ is